

DNA Double Helix and Chromosome Compaction
Astonishing Parallels to T0-Torus Geometry From Molecular
Winding to Highest Information Density

Contents

.1	Introduction: The Packaging Problem	1
	DNA: 2 Meters in 6 μm	1
	T0: Universal Information in Space	2
	The Common Solution	2
.2	The DNA Hierarchy	2
	Level 1: The Double Helix (Molecular)	2
	Level 2: Nucleosomes (Histones)	2
	Level 3: 30-nm Fiber (Solenoid)	3
	Level 4: Higher Loops (~ 300 nm)	3
	Level 5: Condensed Chromatin	3
	Level 6: Metaphase Chromosome (Maximum Compaction)	3
.3	The T0 Hierarchy	3
	Level 1: Fundamental (Sub-Planck)	3
	Level 2: Particles ($\sim 10^{-15}$ m)	4
	Levels 3–6: Scale-Invariant Hierarchy	4
.4	Ten Parallels	4
	Parallel 1: Double Helix as Basic Structure	4
	DNA	4
	T0	4
	Parallel 2: Winding Numbers Determine Properties	5
	DNA	5
	T0	5
	Parallel 3: Hierarchical Compaction	5
	DNA	5
	T0	6
	Parallel 4: Toroidal Geometry	6
	DNA	6
	T0	6
	Parallel 5: Singularity Avoidance	6
	DNA	6
	T0	7
	Parallel 6: Information Maximization	7
	DNA	7
	T0	7
	Parallel 7: Compression Factor	8
	DNA	8
	T0	8
	Parallel 8: Fractal Self-Similarity	8
	DNA	8
	T0	8
	Parallel 9: Topological Stability	9
	DNA	9

	T0	9
	Parallel 10: Dynamic Unfolding	9
	DNA	9
	T0	9
.5	Why These Parallels?	10
	Universal Optimization Problem	10
	Mathematical Reasons	10
	Evolution vs. Fundamentality	11
.6	Quantitative Comparisons	11
	Compression Factors	11
	Hierarchy Levels	11
	Characteristic Lengths	12
.7	Conclusion	12
	Summary Interpretation	12

Correction notes

Corrections from the calculation review (30 Sept. 2026). The following places are corrected or annotated in the text; each carries a dated note there with the previous value:

- Seventh parallel and table of hierarchy levels: T0 7500^{60} (60 levels) $\rightarrow 10^{61} \approx 7500^{16}$ (about 16 levels); $7500^{60} \approx 10^{232}$.
- Section T0 "60+ Hierarchy Levels" and third parallel: $60+ \rightarrow \sim 16\text{--}17$ levels.
- Parallel 7 (DNA) and table of compression factors: $2\text{ m}/6\text{ }\mu\text{m} \approx 333,000$ as "compression factor" $\rightarrow$ genome/chromosome; compaction per chromosome $\sim 10^4$.
- Introduction (DNA in the nucleus): "compression factor $\sim 10,000$ " for 2 m in $6\text{ }\mu\text{m} \rightarrow$ linear ratio $\approx 3.3 \times 10^5$; $\sim 10^4$ applies per chromosome.

Abstract

This paper examines the structural parallels between the DNA double helix, its hierarchical compaction into chromosomes, and the 4D torsional structure of T0 theory. The analysis shows: Both systems can be described by **the same geometric strategy – double helices winding around tori, which in turn fold hierarchically** – to store maximum information in minimum volume. The study describes **ten parallels**: (1) **Double helix as basic structure**, (2) **Winding numbers determine properties**, (3) **Hierarchical compaction across levels**, (4) **Toroidal geometry at each level**, (5) **Singularity avoidance through minimum radii**, (6) **Information maximization with volume minimization**, (7) **10,000-fold compression without loss**, (8) **Fractal self-similarity**, (9) **Topological stability**, (10) **Dynamic unfolding when needed**. The parallels suggest that DNA compaction is a **biological solution to a geometric problem** that, in the T0 description, also structures physics across many scales; they are to be read as a structural analogy, not as proof of a common mechanism.

Status markers. Statements marked **[S]** are *speculative*: a hint or conjecture that has not yet been derived or numerically checked in the corpus. Here this applies in particular to the cosmic scales; the current corpus deliberately sets the cosmic sector aside, since the Standard Model does not derive it either (P39).

.1 Introduction: The Packaging Problem

DNA: 2 Meters in 6 μm

Every human cell faces a considerable geometric problem:

How does one pack ~2 meters of DNA into a nucleus of ~6 μm diameter?

This corresponds to a linear ratio of $2\text{ m}/6\text{ }\mu\text{m} \approx 3.3 \times 10^5$; the compaction per chromosome is **~10,000**. (Corrected on 30 Sept. 2026: previously "compression factor of ~10,000" for 2 m in 6 μm – $2/(6 \times 10^{-6}) = 3.33 \times 10^5$; ~ 10^4 is the compaction per chromosome, cf. the note in Parallel 7.)

T0: Universal Information in Space

T0 theory faces an analogous problem:

How does one encode maximum physical information in finite space without singularities?

The Common Solution

The Universal Principle

Both can be described by the same geometric strategy:

Double helices → wind around **tori** → which **fold hierarchically** → and **dynamically unfold** when needed

This strategy is an **efficient solution for information storage**.

.2 The DNA Hierarchy

Level 1: The Double Helix (Molecular)

Structure:

- Two antiparallel polynucleotide strands
- Right-handed helix
- Turn: 360° per 10.5 base pairs
- Diameter: ~2 nm
- Pitch: ~3.4 nm per turn

Geometry:

$$\text{Winding number } w = \frac{n_{\text{base pairs}}}{10.5} \approx \frac{L}{3.4\text{ nm}} \quad (1)$$

Level 2: Nucleosomes (Histones)

Structure:

- DNA wraps 1.65 times around histone octamer
- Histone core diameter: ~11 nm

- 147 base pairs per nucleosome
- "Beads on a string"

Compression: ~6-fold

Geometry – torus:

$$R_{\text{Histone}} \approx 5.5 \text{ nm}, \quad r_{\text{DNA}} \approx 1 \text{ nm}$$

(2)

The DNA forms a **toroidal loop** around the histone core.

Level 3: 30-nm Fiber (Solenoid)

Structure:

- Nucleosome chain folds into **solenoid**
- 6 nucleosomes per turn
- Diameter: ~30 nm
- "Fiber of fibers"

Compression: ~40-fold (cumulative)

Geometry – helix of tori

Level 4: Higher Loops (~300 nm)

Structure:

- 30-nm fiber forms loops
 - Loops attached to protein scaffold
 - Diameter: ~300 nm
- Compression:** ~400-fold (cumulative)

Level 5: Condensed Chromatin

Structure:

- Further folding of loop domains
 - Diameter: ~700 nm
- Compression:** ~1,000-fold (cumulative)

Level 6: Metaphase Chromosome (Maximum Compaction)

Structure:

- Highest condensation during cell division
 - Length: ~1–10 μm
 - Diameter: ~1 μm
 - X-shaped structure (two sister chromatids)
- Compression:** ~10,000-fold.

2 meters DNA $\rightarrow$ 6 μm nucleus

.3 The T0 Hierarchy

Level 1: Fundamental (Sub-Planck)

Structure: 4D torsional crystal

- Double loop – analogous to DNA double strand
- Toroidal + poloidal circulation
- Winding number $w = n_\phi / n_\theta$
- Minimum radius: $r_{\min} = 21\ell_P$

Level 2: Particles ($\sim 10^{-15}$ m)

Structure: Elementary particles as torus resonances

- Electrons, quarks = stable windings
- Toroidal structure on Compton scale
- Spin from winding number

Levels 3–6: Scale-Invariant Hierarchy

Further torus structures on all scales up to cosmic **[S]**:

- Atoms $\sim 10^{-10}$ m
- Planets $\sim 10^6$ m
- Stars $\sim 10^9$ m
- Galaxies $\sim 10^{20}$ m

Compression: ~ 60 orders of magnitude with $D_f = 3 - \xi$.

.4 Ten Parallels

Parallel 1: Double Helix as Basic Structure

DNA

The **double helix** is the fundamental structure:

- Two strands wound around each other
- Right-handed
- Complementary (A-T, G-C)
- Stability through **both** strands

T0

The electron model of Williamson & van der Mark (1997) describes a **double loop** that can be read as a double helix; the T0 description takes up this structure. The parallel is a classification made in this document, not a statement by the two authors:

- Two circulations: toroidal + poloidal
- Circularly polarized field
- Winding over Compton wavelength λ_C

- Stability through **both** circulations

First Parallel

Double Circulation / Double Helix

Both use **two intertwined components**:

- DNA: Two nucleotide strands
- T0: Toroidal + poloidal flow

In both cases the **factor 2** contributes to stability.

Parallel 2: Winding Numbers Determine Properties

DNA

The **number of turns** determines:

- Helix length
- Number of base pairs
- Topological properties (linking number)
- Supercoiling behavior

Example: Plasmid with 4,000 base pairs has ~380 helix turns

T0

The **winding number** $w = n_\phi/n_\theta$ determines:

- Spin: $w = 1/2 \rightarrow$ fermions
- Spin: $w = 1 \rightarrow$ bosons
- Charge from flux quantization
- Mass from resonance

Second Parallel

Winding Number = Quantum Number

DNA	T0
Number of turns determines length	Winding number determines spin
Linking number topological	Winding number topological
Supercoiling energy	Field energy

Parallel 3: Hierarchical Compaction

DNA

6 Hierarchy Levels:



T0

~**16–17 Hierarchy Levels** (at a factor of ~7500 per level): (Corrected on 30 Sept. 2026: previously "60+ Hierarchy Levels" – 10^{-39} to 10^{26} m are 65 orders of magnitude, which at 7500 per level means about 17 levels; 60+ is the number of orders of magnitude.)

From Sub-Planck (10^{-39} m) to Cosmic (10^{26} m)

Third Parallel

Both use **hierarchical folding across multiple scales**:

DNA: 6 levels, 10,000-fold compression

T0: ~16–17 levels, self-similar with $D_f = 3 - \xi$

(Corrected on 30 Sept. 2026: previously "T0: 60+ levels" – see the note above.)

Parallel 4: Toroidal Geometry

DNA

Torus at every level:

Level 2 (Nucleosomes): DNA wraps **1.65 times** around histone core

$$\text{Torus : } R = 5.5 \text{ nm, } r = 1 \text{ nm} \quad (3)$$

Level 3 (Solenoid): Nucleosome chain forms **helix** (torus-like)

Level 4+: Loop domains attached to central axis = **toroidal arrangement**

T0

Torus on every scale (cosmic level **[S]**):

- Sub-Planck: Fundamental 4D torus
- Particles: Torus resonances
- Macro: Magnetic fields, plasmatoroids
- Cosmic: Galactic spirals, cosmic web

Fourth Parallel

The torus as a recurring geometry

Why? Because it:

- Is closed (no boundaries)
- Enables two independent circulations

- Stores energy/information efficiently
- Is topologically stable (genus = 1)

Parallel 5: Singularity Avoidance

DNA

Minimum radii prevent collapse:

- DNA helix cannot go below ~ 1 nm radius
 - Nucleosomes have fixed core diameter
 - 30-nm fiber has minimum bending
 - Too strong compression $\rightarrow$ DNA damage
- Reason:** Steric hindrance, Van der Waals radii, hydrogen bonds

T0

Minimum torus radius:

$$r_{\min} = 21\ell_p \approx 3.4 \times 10^{-34} \text{ m} \quad (4)$$

Reason: Fractal dimension $D_f = 3 - \xi$ prevents singularity

Fifth Parallel

Both have fundamental lower limit

	DNA	T0
Minimum radius	~ 1 nm	$21\ell_p$
Cause	Chemical	Geometrical
Consequence	DNA stability	No singularity

Parallel 6: Information Maximization

DNA

Problem: 3 billion base pairs of information in $\sim 6 \mu\text{m}$

Solution: Hierarchical folding

Result:

- Information density: $\sim 10^9$ bits / μm^3
- One of the highest known information densities in biology
- Access when needed through local unfolding

T0

Problem: Maximum physical information in finite space

Solution: Fractal torus folding

Result:

- Holographic principle: Information on surface

- Folding maximizes surface area
- Torus has maximum surface area for given volume

Sixth Parallel

Both maximize $\frac{\text{Information}}{\text{Volume}}$

The folding is the **solution to an optimization problem.**

Parallel 7: Compression Factor

DNA

Quantitative:

Stretched DNA : $\sim 2 \text{ m}$ (5)

Chromosome : $\sim 6 \mu\text{m}$ (6)

Genome/chromosome : $\frac{2 \text{ m}}{6 \mu\text{m}} \approx 333,000$ (7)

(Corrected on 30 Sept. 2026: previously "Compression factor" – $2 \text{ m}/6 \mu\text{m}$ compares the whole genome with a single chromosome; the compaction per chromosome is $\sim 10^4$, as stated in the following sentence.)

Considering diameter: **$\sim 10,000$ -fold**

T0

Quantitative:

Planck scale : 10^{-35} m (8)

Hubble scale : 10^{26} m (9)

Orders of magnitude : 61 (10)

With $\xi = 1.33 \times 10^{-4}$: Scaling factor $\sim 1/\xi \approx 7500$ per level.

Seventh Parallel

Both achieve enormous compression without information loss

DNA: 10,000-fold (6 levels)

T0: $10^{61} \approx 7500^{16}$ (about 16 levels)

(Corrected on 30 Sept. 2026: previously "T0: 7500^{60} (60 levels)" – $7500^{60} \approx 10^{232}$, not 10^{61} ; 10^{61} with 7500 per level corresponds to about 16 levels.)

Parallel 8: Fractal Self-Similarity

DNA

Self-similar structure:

- Helix (Level 1) $\rightarrow$ winds into solenoid (helix of helices, Level 3)
- Nucleosomes (tori, Level 2) $\rightarrow$ arranged on helix (Level 3)
- 30-nm fiber $\rightarrow$ folds into loops (Level 4) $\rightarrow$ into chromatin (Level 5)

Each level is a folded version of the previous one.

T0**Strict self-similarity:**

$$\frac{R_{\text{Level } n+1}}{R_{\text{Level } n}} = \frac{1}{\xi} \approx 7500 \quad (11)$$

In this model, the ratio R/r remains constant across scales.

Eighth Parallel**Fractal repetition of the same pattern**

DNA: Qualitatively self-similar (helix → solenoid → loops)

T0: Quantitatively self-similar ($D_f = 3 - \xi$, fixed scaling ratio)

Parallel 9: Topological Stability**DNA****Topological invariants:**

- **Linking number** (Lk): Number of intertwinings
- **Twist** (Tw): Local turns
- **Writhe** (Wr): Supercoiling

Fundamental relationship:

$$\text{Lk} = \text{Tw} + \text{Wr} \quad (12)$$

These numbers are **topologically invariant** – they change only through cutting.

T0**Topological quantum numbers:**

- Winding number $w = n_\phi / n_\theta$
 - Flux quantization $\Phi = n \cdot h/e$
 - Charge, spin, color charge from topology
- These are **topologically protected** – they change only at phase transitions.

Ninth Parallel**Topological stability**

Both use **topological invariants** for stability:

DNA: Linking number preserves structure

T0: Winding number preserves quantum numbers

Parallel 10: Dynamic Unfolding**DNA****Unfolding when needed:**

- **Transcription:** Local unfolding for RNA polymerase
- **Replication:** Complete unfolding during S-phase
- **Recombination:** Temporary unfolding for repair

- **Regulation:** Acetylation → loose structure → accessibility
The compaction is **reversible** and **regulatable**.

T0

Dynamic processes:

- Energy flows in torus variable
- Torsion waves propagate
- Particle creation = excitation
- Phase transitions possible
The structure is **static**, the energy **dynamic**.

Tenth Parallel

Static structure, dynamic processes

	DNA	T0
Structure	Chromosome (static)	Torsion crystal (static)
Dynamics	Local unfolding	Energy flows
Re-versible?	Yes	Yes (excitations)

.5 Why These Parallels?

Universal Optimization Problem

The Fundamental Question

Both biology (DNA) and physics (T0) face a **comparable challenge**:

How does one store maximum information (sequence / physical states) in minimum space without:

- Knotting (topology problems)
- Singularities (infinite energies)
- Information loss (entropy)
- Inaccessibility (must remain readable)

In both cases the answer is: **hierarchical torus folding with double helices**.

Mathematical Reasons

The following general reasons support the parallels:

1. Topology:

- Torus (genus = 1) is simplest non-trivial closed surface

- Enables two independent circulations
- Topologically stable
- **2. Geometry:**
- Helix is natural curve in 3D
- Double helix maximizes stability
- Winding around a torus is a favorable arrangement

3. Information theory:

- Holographic principle: Information on surface
- Folding maximizes surface area
- Hierarchy allows logarithmic compression

Evolution vs. Fundamentality

An Interpretation

Did evolution "discover" torus geometry?

A possible answer [S]:

Evolution may have found this geometry because it is a **very favorable solution** to the information storage problem.

In the T0 description, physics also uses this geometry for fundamental structure.

DNA compaction would then be **not a random biological invention**, but an **expression of a general geometric principle**. Whether torus folding is really the only optimal solution is not proven in this document.

.6 Quantitative Comparisons

Compression Factors

System	From	To	Factor
DNA	~4–5 cm (stretched)	6 μm (chromosome)	~10,000×
T0	10^{-35} m (Sub-Planck)	10^{26} m (cosmic)	10^{61}

Table 1: Compression factors

(Corrected on 30 Sept. 2026: previously DNA from 2 m, factor 333,000× – 2 m is the whole genome; a single chromosome measures on average ~4–5 cm when stretched, and the compaction per chromosome is $\sim 10^4$.)

Hierarchy Levels

System	Levels	Factor/Level	Geometry
DNA	6	$\sim 2-6\times$	Helix + Torus
T0	~ 16	$\sim 7500\times$	Torus + Fractal

Table 2: Hierarchical structure

(Corrected on 30 Sept. 2026: previously T0 "60+" levels – with 7500 per level, about 16 levels span the 10^{61} of the table of compression factors; 60 is the number of orders of magnitude.)

Characteristic Lengths

DNA Level	Length	T0 Analog	Length
Double helix	2 nm	Sub-Planck	10^{-39} m
Nucleosome	11 nm	Particle	10^{-15} m
30-nm fiber	30 nm	Atom	10^{-10} m
Loop	300 nm	Molecule	10^{-9} m
Chromatin	700 nm	Macro	10^0 m
Chromosome	1 μm	Cosmic	10^{26} m

Table 3: Scale comparison (qualitative)

.7 Conclusion

Main Result

DNA compaction and T0 torus geometry show **ten structural parallels**:

1. Double helix / Double circulation
2. Winding numbers = quantum numbers
3. Hierarchical compaction
4. Toroidal geometry at each level
5. Singularity avoidance through minimum radius
6. Information maximization
7. Enormous compression factors
8. Fractal self-similarity
9. Topological stability
10. Dynamic unfolding

The parallels point to a **common geometric solution** for information storage; whether a common principle lies behind it remains an open question.

Summary Interpretation

Interpretation [S]

**In this reading, biology and physics use the same geometry
because it is a particularly favorable solution.**

DNA compaction can be read as a **biological expression** of the same **geometric principle** that, in the T0 description, also:

- Structures brain gyri
- Forms elementary particles
- Organizes the universe

In this interpretation, nature uses **the same solution on many scales** and **in different domains**:

Double helices → Tori → Hierarchical folding

(13)

This is a **common answer** to the problem:

Maximize information, minimize space, avoid singularities.